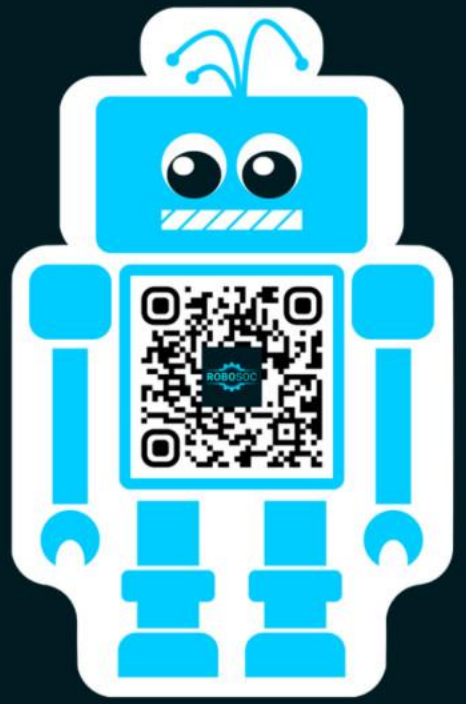

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Workshop 5:

ESP32 & Robots



What is an ESP32?



INPUT

buttons, switches, sensors



BRAIN

ESP32



OUTPUT

LEDs, screens, buzzers, motors

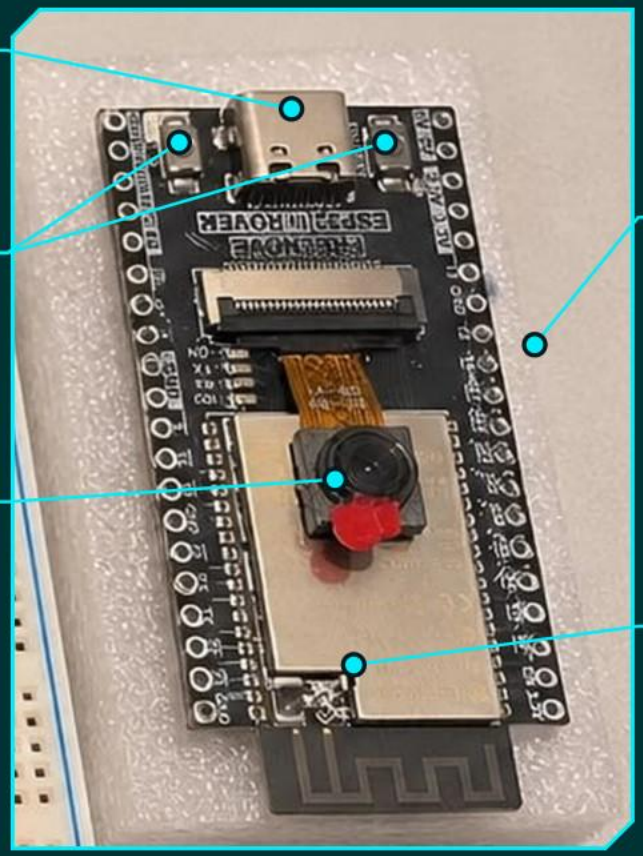
- A microcontroller: a small computer on a chip that runs one program over and over
- Same job as last workshop's Arduino Uno, but a lot faster
- Wi-Fi and Bluetooth built in
- Found in smart plugs, fitness trackers, weather stations and drones

ESP32 board

USB-C
power and code upload

EN · BOOT buttons
restart · hold if upload sticks

Camera
not used today



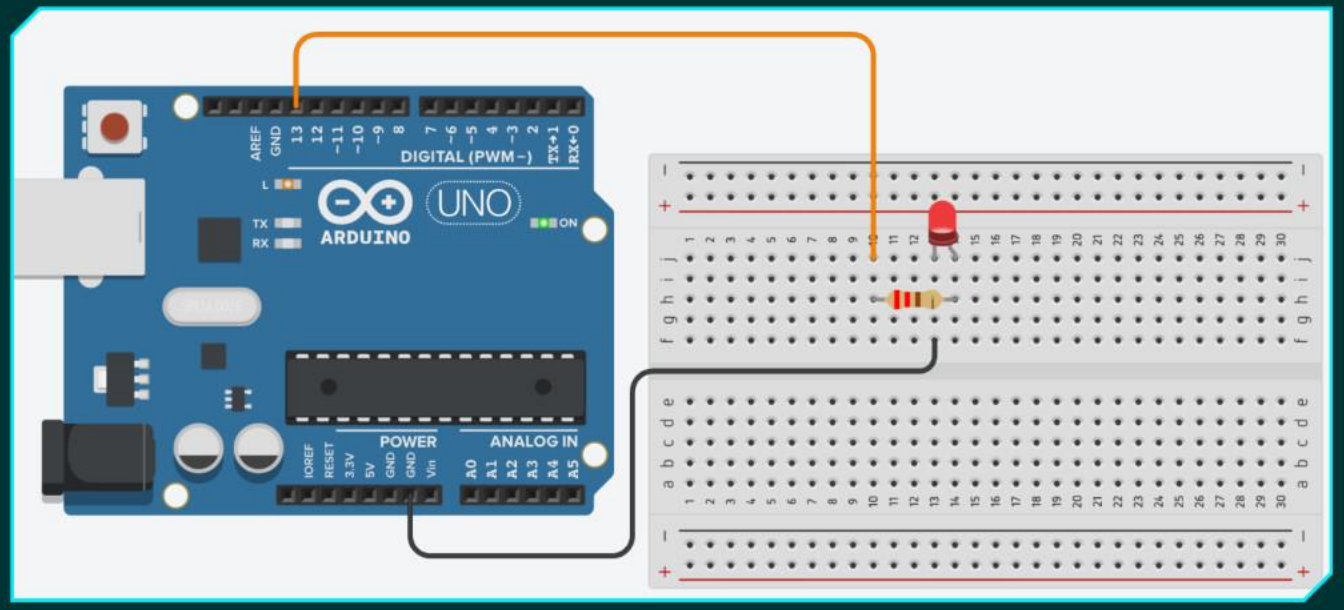
Header pins
named by GPIO number

ESP32 module
the chip, 3.3 V logic

Pin rules 3.3 V only · pins 34–39 are inputs only · avoid 0, 2, 6–11, 12 and 15

Activity 1: Blink an LED

Pin 13 → 220 Ω resistor → LED → GND



Simulated in Tinkercad: same circuit on an Arduino Uno, also pin 13

- Pin 13 to the resistor
- Resistor to the LED's long leg
- Short leg to GND
- Upload `blink.ino`

Try it in Tinkercad

Activity 1: The code

```
// RoboSoc Workshop 5: Activity 1, blink an LED

const int LED_PIN = 13;

void setup() {
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(115200);
}

void loop() {
  digitalWrite(LED_PIN, HIGH); // on
  Serial.println("on");
  delay(500);
  digitalWrite(LED_PIN, LOW);  // off
  Serial.println("off");
  delay(500);
}
```

- Serial Monitor at 115200: on, off, on, off
- LED not on? Turn it round

Have a go!

- Make the LED blink faster
- Blink SOS in morse code (· · · – – – · · ·)
- Add a second LED on pin 14 that turns on when the first one turns off
- Add a push button on pin 32 so the LED only blinks while you hold it

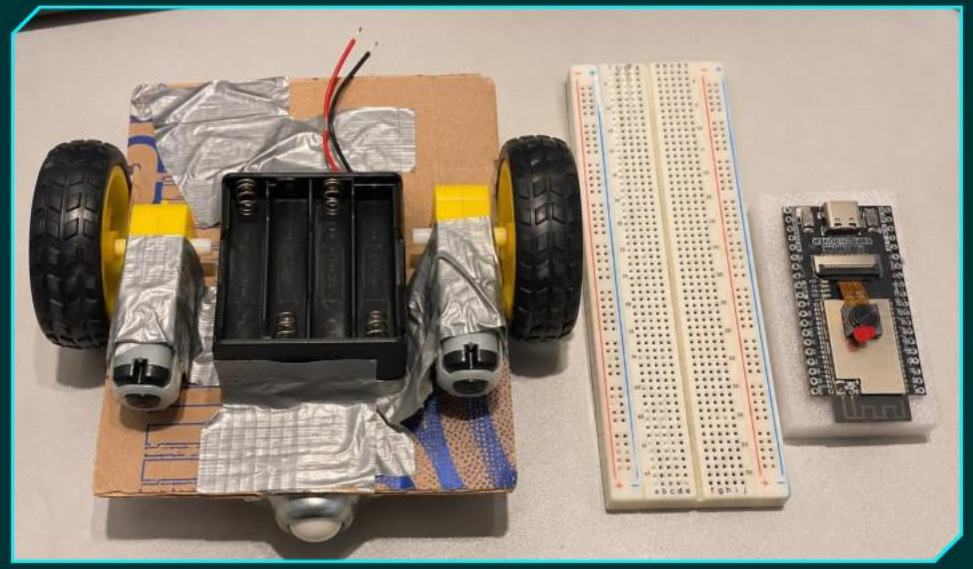
What are we building?

INPUT
ultrasonic sensor

BRAIN
ESP32

OUTPUT
motor driver + wheels

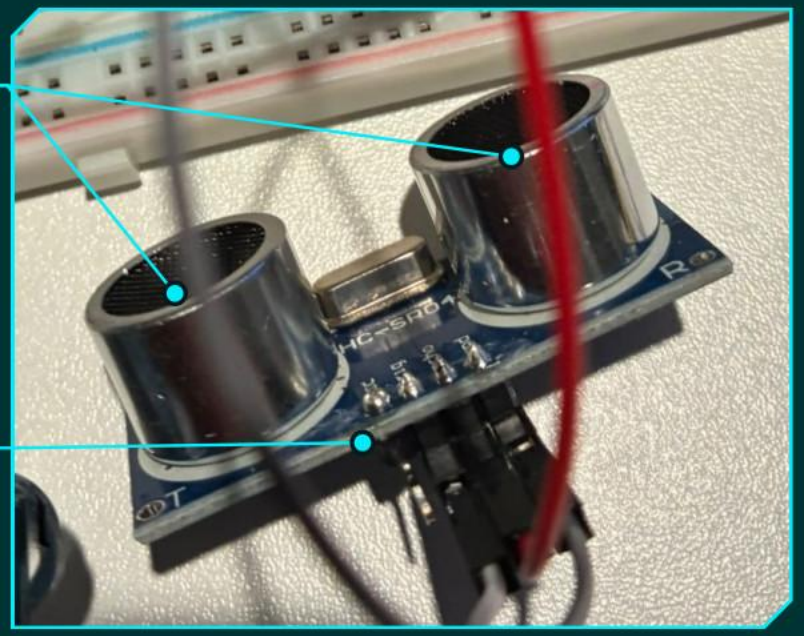
Closer than 20 cm: stop, reverse, turn.

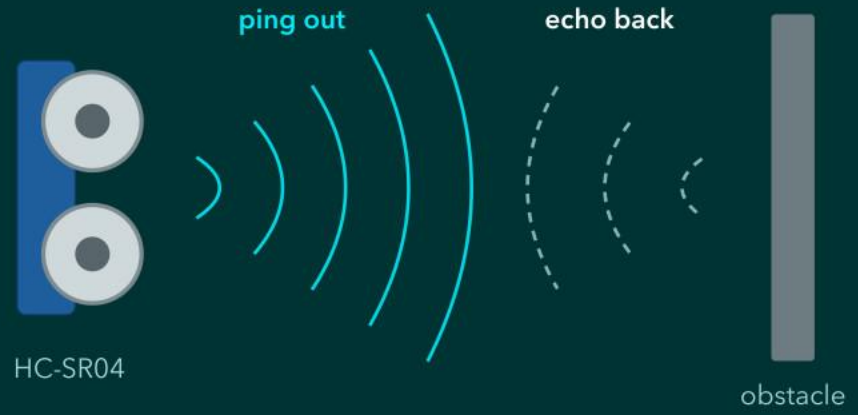


HC-SR04 ultrasonic sensor

Transducers
one sends, one listens

Pins
VCC · TRIG · ECHO · GND

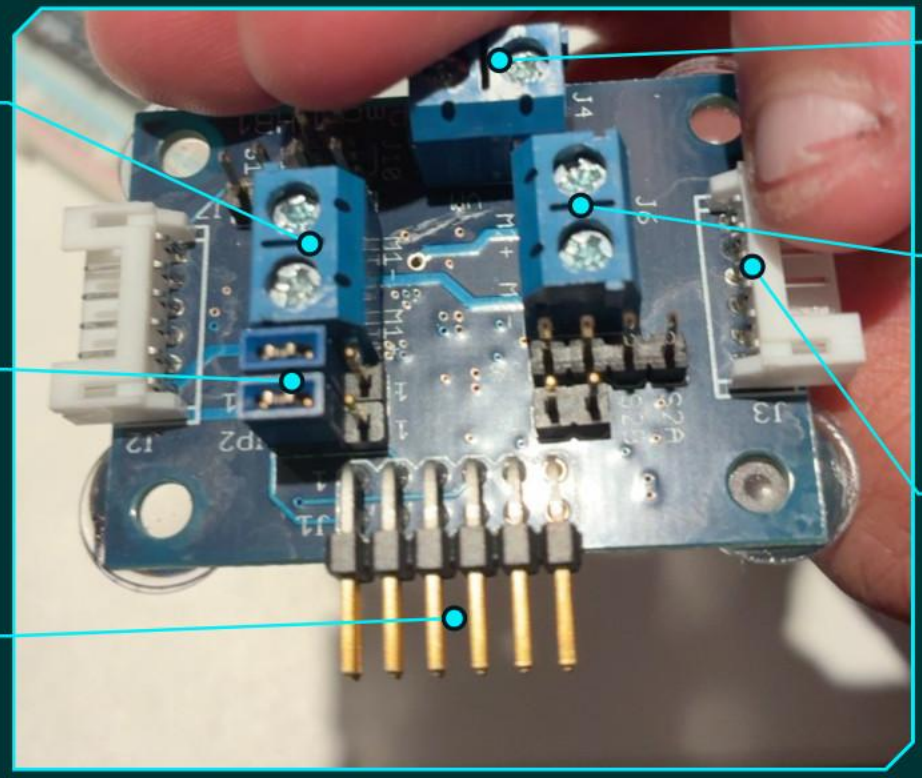





```
cm = echo µs × 0.0343 ÷ 2
```

range about 2 cm to 4 m

Motor driver: Pmod DHB1



J5
left motor

JP1 · JP2
both on position 3

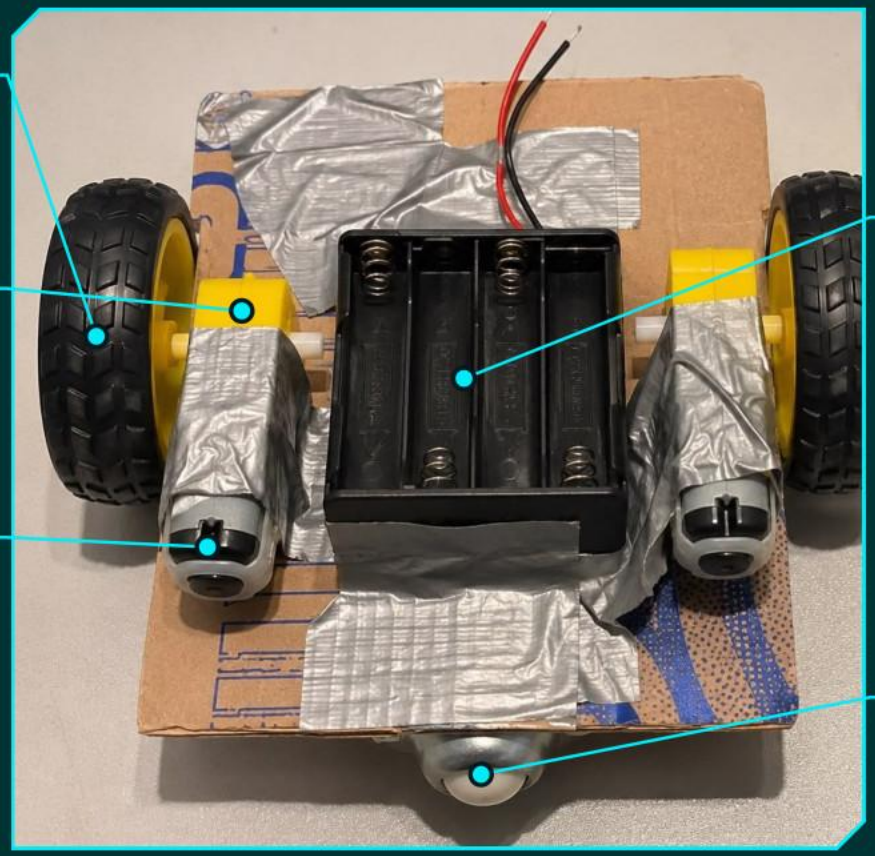
J1
control pins from the ESP32

J4
batteries in

J6
right motor

J3 · J2
white sockets, not used

Motors, wheels and caster



Wheel
one per motor

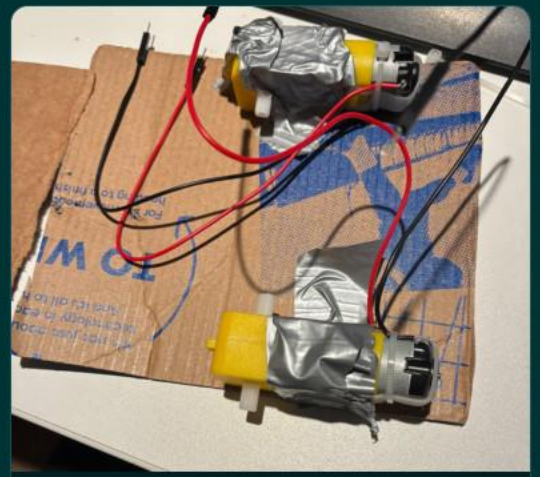
Yellow gearbox
trades speed for strength

DC motor

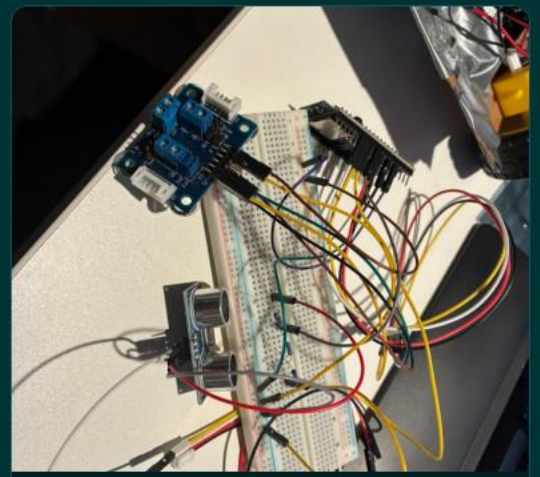
Battery holder
4 × AA

Ball caster
underneath, at the front

Build it with the guide



1
Building the chassis
Is already partially built for some of you



2
Electrical connections
`steps 7-22`



3
Final assembly
`steps 23-30`

Build guide
scan to open the PDF

Testing, troubleshooting, Tinkercad simulations and challenges are all in the guide
`steps 1-6`

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COMPETITION TEAM SIGN-UP



REMINDER!

YOU MUST PURCHASE A MEMBERSHIP IN ORDER TO ATTEND WORKSHOPS AND TAKE PART IN COMPETITIONS!!!

YOU WILL ALSO GET PRIORITY FOR SOCIAL EVENTS AND BADMINTON / VOLLEYBALL TEAM

ONLY £5 ON OUR SU PAGE

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